

# **Forecasting Crude Oil Prices – Step 2 Analysis**

## **Trend Analysis**

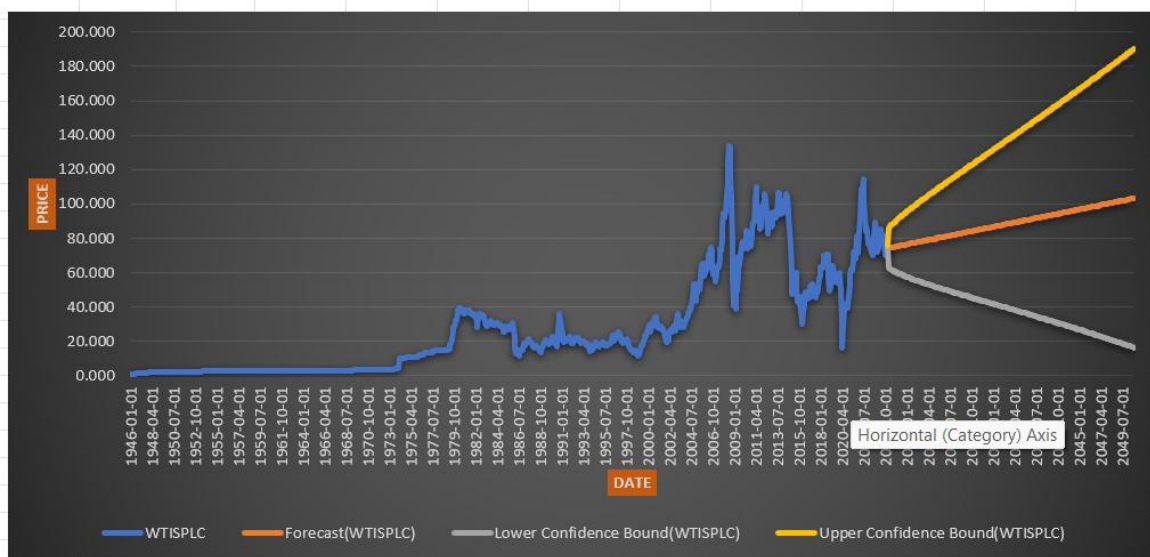
The forecast results indicate that WTI Spot Crude Oil prices show a long-term upward trend. Historical data reveals relatively low prices in the early years, followed by gradual increases. The forecast line continues this growth pattern into the future, suggesting stable long-term price appreciation.

## **Seasonality Patterns**

The forecast statistics show  $\text{Gamma} = 0$ , meaning no significant seasonal patterns were detected. This implies that crude oil prices do not follow a strong yearly cycle but are instead driven by broader economic and market trends.

## **Future Price Fluctuations**

The confidence intervals widen over time, reflecting increasing uncertainty in long-term predictions. For example, in January 2050, the forecasted price is approximately 100 USD per barrel, with a lower bound near 80 USD and an upper bound near 120 USD. This range highlights potential volatility in future oil markets.



## Forecast Statistics Interpretation

- **Alpha (0.25):** The model gives more weight to historical data, adapting slowly to new changes.
- **Beta (0.00):** No additional trend smoothing, indicating a stable trend component.
- **Gamma (0.00):** No seasonal adjustment, confirming the absence of repeating cycles.
- **MASE (7.41) & RMSE (9.45):** These error values show moderate forecast error, meaning results should be interpreted cautiously.
- **SMAPE (0.11 or 11%):** Indicates relatively good accuracy, as the average percentage error is low.
- **MAE (6.98):** The average absolute error is about 7 units, consistent with the RMSE.

| Statistic | Value |
|-----------|-------|
| Alpha     | 0.25  |
| Beta      | 0.00  |
| Gamma     | 0.00  |
| MASE      | 7.41  |
| SMAPE     | 0.11  |
| MAE       | 6.98  |
| RMSE      | 9.45  |

**Conclusion** Overall, the forecast suggests a steady upward trend in crude oil prices with no strong seasonal effects. While the forecast accuracy is acceptable (SMAPE = 11%), the widening confidence intervals highlight the uncertainty of long-term predictions. Decision-makers should consider both optimistic and pessimistic scenarios when planning future strategies.